

Homework 6

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1 Python

1. The OLS coefficient for the impact of mpg on the price of a car is -22.21.
2. The OLS regression likely suffers from omitted variable bias as price of a car may be influenced by other factors. Eg: Horsepower, brand reputation, technology, etc.
3. The second-stage coefficients are as follows:
 - (a) See Table 1.
 - (b) See Table 1.
 - (c) See Table 1.
 - (d) For the exclusion restriction to hold, the instruments must influence the car price through mpg only. For these instruments, that does not seem reasonable as both weight and height of a car may influence the price directly.
 - (e) While the estimated coefficients on mpg are similar for (a) and (b), they are very different for (c). This may be due to the fact that height is a weak instrument for mpg as is evident from the first stage F-statistic.
4. The standard errors in Table 2 are heteroskedasticity robust.

2 STATA

1. See Table 3.
2. The 5% critical value is 37.41 for the F-statistic 78.36 implying weight is not a weak instrument for mpg.

3 Tables

Table 1: By-hand IV results

	weight	$weight^2$	height
$\hat{mpg}$	150.43 (62.16)	157.06 (62.02)	10165.74 (26559.83)
Car	-4676.09 (574.37)	-4732.67 (573.29)	-90156.39 (226687.35)
F -stat	75.46	75.77	0.00

Notes: This table reports the second stage results of three 2SLS regressions. Column (1) estimates estimates the regression with weight as an IV for mpg. Column (2) uses $weight^2$ as an IV while Column (3) uses car height. First-stage F-statistic for each excluded instrument is reported at the bottom of the table. Standard errors in parantheses.

Table 2: IV GMM results

	Weight
Car	-4676.09 (589.70)
Mpg	150.43 (63.05)

Notes: This table reports the IV GMM results upon regressing price on mpg and car type with weight as an IV for mpg. Standard errors in parantheses.

Table 3: IV LIML results	
VARIABLES	(1) price
mpg	150.4** (63.05)
car	-4,676*** (589.7)
Constant	17,628*** (1,773)
Observations	1,000
R-squared	0.104
Robust standard errors in parentheses	
*** p<0.01, ** p<0.05, * p<0.1	
<i>Notes: This table reports the IV LIML results upon regressing price on mpg and car type with weight as an IV for mpg. Standard errors in parantheses.</i>	